

MOHAMAD ALI SABER

Eram. Shiraz, Fars, Iran

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Education

University of Tehran

2019-2023

M.Sc. in Electrical Engineering (Digital Electronic Systems)

Tehran, Iran

- **GPA:** 17.47/20
- **Supervisor:** Dr. Zainolabedin Navabi
- **Thesis:** : Implementation of Near Data Processing Based System for CNN Applications

Shiraz University

2014-2019

B.Sc. in Electrical Engineering

Shiraz, Iran

- **GPA:** 15.74/20

Research Interests

- Machine Learning
- Artificial Intelligence
- Computer Architecture
- Computer Arithmetic
- Parallel Processing
- Embedded Systems

Publications

- T. Vasei, **M. Ali Saber**, A. Nahvy, and Z. Navabi, “An efficient RTL design for a wearable brain-computer interface,” IET Computers Digital Techniques, 2024
- R. Sadeghi, E. Akbari, and **M. Ali Saber**, “On-chip training of crosstalk predictors to fit uncertainties,” 2022 IEEE European Test Symposium (ETS), 2022. doi: 10.1109/ETS54262.2022.9810362.
- M. S. Roodsari, **M. Ali Saber**, and Z. Navabi, “Diba: N-dimensional bitslice architecture for lstm implementation,” 2020 23rd International Symposium on Design and Diagnostics of Electronic Circuits Systems (DDECS), 2020. doi: 10.1109/DDECS50862.2020.9095614 .
- **M. Ali Saber** and Z. Navabi, “NEMESIS: A 3D Memory-Based Near-Memory Processing Architecture for CNN Acceleration,” (Under Review), 2026

Experience

Teaching Experience:

- * **Teacher Assistance of Reinforcement Learning** Dr. Rastegar. Spring 2026
- * **Teacher Assistance of Digital Logic Design** Dr. Navabi; Dr.Yasrebi. Spring 2016-2026
- * **Teacher Assistance of Digital Logic Design Lab** Dr. Navabi; Dr.Yasrebi. Spring and Fall 2017-2026
- * **Teacher Assistance of Microprocessor** Dr. Yasrebi. Spring 2018-2026

Work Experience:

- * **junior Developer:** junior FPGA(VHDL) Developer in Hamrah-e-Aval. Spring and Summer 2022
- * **junior Developer:** junior FPGA(VHDL) Developer in PPAS-Co. Spring and Summer 2021
- * **Intern:** Researcher in Image and Signal Processing Lab. under supervision of Dr. Yasrebi. Summer 2018
- * **Freelancer:** More than 100 academic and industrial projects. 2020-2026

Awards and Honors

- Summer 2019: Ranked 44th among more than 20,000 participants in the national university entrance exam for master's program in electrical engineering.
- Summer 2014: Rank 1505th regionally among more than 180,000 participants in the national university entrance exam

Relevant Coursework

- | | |
|---|---|
| * Neural Network and Deep Learning:17.64 out of 20 | * Reinforcement Learning:19.7 out of 20 |
| * Computer Arithmetic:19 out of 20 | * VHDL and SystemC:17.8 out of 20 |
| * Technologies, Circuits and Memory Systems:17.25 out of 20 | * Computer Architecture:17.27 out of 20 |
| * Microprocessor:19.77 out of 20 | * Interconnects:18.53 out of 20 |
| * Digital System Test and Testability:19.3 out of 20 | * Fault Tolerant System:16.9 out of 20 |

Selected Projects

Machine Learning Projects

2020-2026

- Developed and trained Reinforcement Learning (RL) agents for multiple game environments, including Lunar Lander, Bricks, and Post Delivery, to explore and evaluate RL algorithms and training strategies.
- Implemented and trained a variety of Machine Learning models, including Self-Organizing Maps (SOMs), Multi-Layer Perceptrons (MLPs), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Generative Adversarial Networks (GANs), and Large Language Models (LLMs).

Processor and Accelerator Design

2018-2026

- Designed and implemented multiple processor architectures in HDL, including 8085, MIPS (single-cycle and multi-cycle), AVR (multi-cycle and pipelined), and RISC-V cores.
- Designed and implemented hardware accelerators for computationally intensive applications, including CNN inference, Support Vector Machines (SVMs), and matrix multiplication.
- Developed Instruction Set Simulators (ISSs) for several custom processors, including SAYAK, PUNEH, and AFTAB, to support software development and architectural verification.

Tool Development

2018-2026

- Developed an Instruction Set Simulator (ISS) Generator for rapid creation of processor simulation environments.
- Developed a CNN compiler targeting custom accelerator architectures, enabling automated mapping and execution of neural network workloads.

Technical Skills

* Programming Languages:	Python, MATLAB, C/C++, Assembly(AVR, X86, and MIPS)
* Machine Learning Tools:	TensorFlow, Pytorch, Keras, Scikit-Learn
* Hardware Description Languages:	VHDL, Verilog, SystemVerilog
* System Level Modeling Languages:	SystemC, SystemC-AMS, Chisel, HLS
* Hardware:	FPGA, ARM, AVR
* Applications:	Simulink, Xilinx (ISE, Vivado), Quartus, ModelSim, Cadence, Microsoft Office
* Tools:	Git, Latex, Linux, Docker
* Language:	Persian (Native), English (Fluent)

References

Available on Request